Band-1 Weinstein Conjecture: One Contact Setting Probe (Collapse 2.35)

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Band-1 Weinstein — One New Contact Setting (Lit + Statement Lock)

Honesty stamp. This note does **not** claim the Weinstein conjecture in all contact settings. It does **not** claim the Weinstein conjecture in all dimensions for arbitrary contact structures. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. Lit lock only — **no** symplectic numerics / Reeb integration on box.

> **Locked PROVED (dim 3):** every closed oriented contact 3-manifold has a closed Reeb orbit (**Taubes 2007**).
> **Locked PROVED (Band-1 one setting): PS-overtwisted (Plastikstufe)** closed contact manifolds in dimension ≥ 5 — every Reeb field has a **contractible** periodic orbit (**Albers–Hofer** line).
> **Locked OPEN:** general closed contact manifolds of dimension ≥ 5 without PS / subcritical-open-book / contact-type hypotheses — **OPEN**.

Abstract

Weinstein (1979) conjectured that every closed co-oriented contact manifold admits a closed Reeb orbit for every defining contact form. In **dimension 3** the conjecture is completely settled by **Taubes** (Seiberg–Witten / embedded contact homology bridge), after partial results of **Hofer** (overtwisted / $\pi_2 \neq 0$ / S^3) and **Abbas–Cieliebak–Hofer** (planar). In **higher dimensions** many **special** classes are known (Viterbo contact-type hypersurfaces in $\mathbb{R}^{2n}$; **PS-overtwisted** / Plastikstufe; subcritical Stein open books), but the **unrestricted** conjecture remains **open**.

This Band-1 probe locks **one new contact setting** beyond the classical 3D theorem: the **PS-overtwisted** setting in dimension ≥ 5. It does **not** elevate that special theorem to an all-settings claim.

Problem (collapse Band-2 item 2.35 → Band-1 lit+statement)

1. Ledger the Weinstein conjecture; known cases (Hofer / Taubes / higher-dim specials).
2. Lock **one** literature-proved higher-dim setting as the Band-1 claim surface.
3. Name **one remaining open class**.
4. Keep Clay / ToE / GATE-04 / "Weinstein in all settings" off the claim surface.
5. Lit lock enough — no box Reeb numerics.

Canonical statement

Let M^{2n-1} be a closed oriented odd-dimensional manifold and α a **contact form** ($\alpha \wedge (d\alpha)^{n-1}$ a volume form). The **Reeb vector field** R_α is uniquely determined by

$$\iota_{R_\alpha} d\alpha = 0, \qquad \alpha(R_\alpha) = 1.$$

Weinstein conjecture: R_α has at least one periodic orbit. Equivalently: every closed co-oriented contact manifold (M, ξ) admits a closed Reeb orbit for every contact form with $\ker \alpha = \xi$.

Literature ledger (web check; not re-proved on box)

Year	Result	Authors	Status
1979	Conjecture stated (after Rabinowitz / Weinstein hypersurface work)	Weinstein	classical statement
1987	Compact hypersurfaces of contact type in $\mathbb{R}^{2n}$ have a closed characteristic	Viterbo	**proved** (special)
1993	Overtwisted contact 3-manifolds; also $\pi_2 \neq 0$ and S^3 — contractible Reeb orbit	Hofer	**proved** (special 3D)
2005	Planar contact structures in dim 3	Abbas–Cieliebak–Hofer	**proved** (special 3D)
2007	**All** closed oriented contact **3-manifolds**	**Taubes**, *Geom. Topol.* 11	**proved** (complete dim-3)
2006–09	Plastikstufe / **PS-overtwisted** $\Rightarrow$ contractible Reeb orbit (higher dim)	Niederkrüger; **Albers–Hofer**, *CMH*	**proved** — **Band-1 setting**
2012	Open books dim ≥ 5 with binding of restricted contact type in a **subcritical Stein** manifold (incl. subcritical pages)	Albers–Geiges–Zehmisch, arXiv:1210.7947	**proved** (special higher-dim)
—	General closed contact manifolds **dim** ≥ 5	—	**OPEN**

Survey (dim 3). M. Hutchings, *Taubes's proof of the Weinstein conjecture in dimension three*, Bull. Amer. Math. Soc. **47** (2010), 73–125.

What is proved vs open (statement lock)

Literature-proved (cite carefully)

1. **Dimension 3, all cases** — Taubes (2007): every Reeb vector field on a closed oriented contact 3-manifold has a closed orbit. Currently the only known complete proof in all 3D cases; uses Seiberg–Witten Floer theory (Kronheimer–Mrowka) linked to Reeb dynamics / ECH.
2. **Hofer specials (3D)** — overtwisted; $\pi_2(Y) \neq 0$; $Y \simeq S^3$ — contractible orbit (holomorphic curves in the symplectization).
3. **Planar (3D)** — Abbas–Cieliebak–Hofer.
4. **Contact-type hypersurfaces in** $\mathbb{R}^{2n}$ — Viterbo (1987).

5. **Band-1 one new setting: PS-overtwisted (Plastikstufe), dim ≥ 5** — Albers–Hofer: every Reeb field of a PS-overtwisted contact form has a **contractible** periodic orbit. Plastikstufe (Niederkrüger) is the standard higher-dimensional overtwistedness witness; PS structures exist on every closed odd-dimensional manifold that admits some co-oriented contact structure (via connected sum with PS spheres). **This is the collapse-2.35 "one new contact setting" locked here.**
6. **Subcritical open-book specials** — Albers–Geiges–Zehmisch: supporting open books whose binding embeds as restricted contact type in a subcritical Stein manifold $\Rightarrow$ contractible Reeb orbit.
7. **Subcritical Weinstein boundaries** — symplectic homology vanishes $\Rightarrow$ Reeb orbit (Viterbo / Cieliebak line).

Locked OPEN (named remaining class)

1. **General closed contact manifolds of dimension ≥ 5 — OPEN**.
2. **Named remaining open class (this probe): tight** (or otherwise **non-PS-overtwisted**) closed contact **5-manifolds** (and higher) **without** a known reduction to subcritical open books, contact-type hypersurface theorems, or SH-vanishing fillings. No Seiberg–Witten / ECH bridge is known that settles the generic higher-dimensional case.

Explicit non-claims

- Weinstein **not** claimed for **all** contact settings.
- Weinstein **not** claimed for **all** dimensions without hypotheses.
- No Clay; no ToE; **GATE-04 LOCKED**.
- No box numerics presented as evidence.

Why Band-1 vs full conjecture

- **Band-1 (this probe):** honest lit+statement lock for **one** higher-dim contact setting (PS-overtwisted) plus the complete dim-3 theorem as classical backdrop.
- **Band 3–4:** unrestricted Weinstein in dim ≥ 5.
- **Collapse note:** "Many cases known" — reflected honestly as specials + full dim-3; remainder open.

Box note

No Reeb-flow numerics, no contact-form integration, no symplectic homology computation on box. Dash rule: **lit lock is enough**.

Deliverables

- Lab: `experiments/band1/weinstein_contact/` (`literature_ledger.json`, `results.json`, `SUMMARY.txt`, `METHOD.md`, `README.md`, local `next_wave.md`).
- Paper/PDF: `papers/2026-09-11_Band1_Weinstein_contact_setting_probe.{md,pdf}` → `DOWNLOAD_PDFs` + Mac `Downloads/DOWNLOAD_PDFs_2026-09-11/`.
- House `next_wave.md` updated.

References (primary)

1. A. Weinstein, *On the hypotheses of Rabinowitz' periodic orbit theorems*, J. Differential Equations **33** (1979).
2. C. Viterbo, *A proof of Weinstein's conjecture in* $\mathbb{R}^{2n}$, Ann. Inst. H. Poincaré Anal. Non Linéaire **4** (1987).
3. H. Hofer, *Pseudoholomorphic curves in symplectizations...*, Invent. Math. **114** (1993).
4. C. Abbas, K. Cieliebak, H. Hofer, *The Weinstein conjecture for planar contact structures in dimension three*, Comment. Math. Helv. **80** (2005).

5. C.H. Taubes, *The Seiberg–Witten equations and the Weinstein conjecture*, Geom. Topol. **11** (2007), 2117–2202.
6. M. Hutchings, *Taubes's proof of the Weinstein conjecture in dimension three*, Bull. AMS **47** (2010).
7. K. Niederkrüger, *The plastikstufe...*, Algebr. Geom. Topol. **6** (2006).
8. P. Albers, H. Hofer, *On the Weinstein conjecture in higher dimensions*, Comment. Math. Helv. **84** (2009), 429–436.
9. P. Albers, H. Geiges, K. Zehmisch, *Open books and the Weinstein conjecture*, arXiv:1210.7947.